

The Solar System

Introduction

The Solar System is the gravitationally bound system of the Sun and the objects that orbit it. It formed about 4.6 billion years ago from the gravitational collapse of a giant interstellar molecular cloud. The vast majority of the system's mass is in the Sun, with most of the remaining mass contained in the planet Jupiter.

The Sun

The Sun is the star at the center of the Solar System. It is a nearly perfect sphere of hot plasma, and its diameter is about 1.39 million kilometers, which is about 109 times that of Earth. The Sun accounts for about 99.86 percent of the total mass of the Solar System. It generates energy through nuclear fusion, converting hydrogen into helium in its core.

The Planets

There are eight planets in the Solar System. In order of distance from the Sun, they are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. The four inner planets (Mercury, Venus, Earth, and Mars) are called terrestrial planets because they have solid, rocky surfaces. The four outer planets (Jupiter, Saturn, Uranus, and Neptune) are called gas giants because they are much larger and made mostly of hydrogen and helium.

Earth

Earth is the third planet from the Sun and the only astronomical object known to harbor life. About 71 percent of Earth's surface is covered with water. Earth has one natural satellite, the Moon, which is the fifth largest moon in the Solar System. Earth takes approximately 365.25 days to orbit the Sun, and it rotates on its axis once every 24 hours.

Jupiter

Jupiter is the largest planet in the Solar System. It is a gas giant with a mass more than two and a half times that of all the other planets in the Solar System combined. Jupiter is known for its Great Red Spot, a giant storm that has been raging for hundreds of years. Jupiter has at least 95 known moons, the largest of which are Io, Europa, Ganymede, and Callisto.

Saturn

Saturn is the second largest planet in the Solar System and is famous for its extensive ring system, made mostly of ice particles with a smaller amount of rocky debris and dust. Saturn has 83 confirmed moons, with Titan being the largest. Titan is the only moon in the Solar System with a dense atmosphere.

Dwarf Planets and Other Objects

Besides the eight main planets, the Solar System contains several dwarf planets, including Pluto, Eris, Haumea, Makemake, and Ceres. The Solar System also contains the asteroid belt, located between the orbits of Mars and Jupiter, and the Kuiper Belt, a region beyond Neptune filled with icy bodies.